Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 421: Software Engineering

Batch: 23rd | Semester: Spring 2026
Date: 01/09/2026 | Final Examination | Time: 2 Hours
Total Marks : 40

Answer any **five** questions: $5 \times 8 = 40$

1. (a) Briefly explain seven distinct tasks of requirements engineering. 4
   (b) Describe the steps required to establish the groundwork for understanding of software requirements. 4

2. (a) Draw a diagram for translating the Requirements Model into the Design Model. 4
   (b) List software quality guidelines and attributes. 4

3. (a) How do we assess the quality of a software design? 4
   (b) Write generic task set for design. 4

4. (a) Explain the golden rules for user interface design. 4
   (b) What is interface analysis? Explain. 4

5. (a) Draw the Quality requirements tree for WebApp. 4
   (b) List and explain the WebApp design quality. 4

6. (a) Illustrate a design pyramid for WebApp. 4
   (b) Why aesthetic design also called graphic design? Explain. 4

7. (a) Briefly explain the software quality dilemma. 4
   (b) Explain four broad activities to achieving software quality. 4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE423: Simulation and Modeling

Batch: 23rd | Semester: Spring 2026

Date: 03-09-2026 | **Final Examination** | Time: 2 Hours

Total Marks: 40

Answer any **five** questions of the followings. $5 \times 8 = 40$

1. (a) Explain the types of simulations with respect to output analysis with example. 4

   (b) Explain Inverse Transform Method for generating random variates with example. 4

2. (a) What is input modeling in simulation? Why is input modelling important in simulation? 4

   (b) An online shopping website wants to study its order-processing system. Orders arrive at different times, enter a processing queue, are processed by available servers, and then leave the system. 4
   Why is discrete-event simulation suitable for this system? Mention three important events that could be modeled.

3. (a) When is a stochastic process called a Continuous-Time Markov Chain? 3

   (b) A city wants to improve the timing of traffic signals at a busy intersection. Before making changes to the real intersection, engineers create a computer model containing vehicles, traffic signals, arrival rates, and road capacity. 5
   Explain how modeling and simulation can help solve this problem. What parameters could be measured to evaluate the proposed solution?

4. (a) List the issues that are considered in an accreditation decision. 3

   (b) Draw the diagram for event scheduling scheme in computer simulation. 5

5. (a) What is Monte Carlo Simulation? What kind of input data is required for performing Monte Carlo simulations? 3

   (b) Use Monte Carlo Simulation to approximate the value of $\int_2^5 x^2 dx$ 5
   R1 → 0.21, 0.11, 0.71, 0.65, 0.41, 0.35, 0.17
   R2 → 0.96, 0.82, 0.56, 0.68, 0.58, 0.40, 0.82

6. Generate a sequence of 5 two-digit random numbers by employing the following methods:

   (i) Mixed Congruential Method 2

   (ii) Multiplicative Congruential Method 2

   (iii) Additive Congruential Method- 2
   The recursive equartion for the mixed congruential method is-
   $R_{i+1} = (21r_i + 53)$ (module 100) and seed $r_0 = 46$

   (iv) Midsquare Method (using same seed) 2

7. A sample of 100 arrivals of a customer at a retail sales depot is according to the following distribution: 8

Time between Arrival (min.)	Frequency
0.5	2
1.0	6
1.5	10
2.0	25
2.5	20
3.0	14
3.5	10
4.0	7
4.5	4
5.0	2

A study of the time required to service customers by adding up the bills, receiving payments and placing packages, yields the following distribution:

Time between Arrival (min.)	Frequency
0.5	12
1.0	21
1.5	36
2.0	19
2.5	7
3.0	5

Estimate the average percentage of customers waiting in time and average percentage of idle time of the server, by simulation, for the next 7 arrivals.
Random number for inter-arrival: 71, 03, 88, 11, 94, 74, 59
Random number for service: 41, 20, 26, 34, 70, 07, 25

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE469 : Digital Image Processing

Batch: 23rd

Semester: Spring 2026

Date: 10/09/2026 **Final Examination** Time: 2 Hours

Total Marks: 40

Answer any **five** questions. $5 \times 8 = 40$

1. (a) Define data redundancy. Mention different types of data redundancies. 2

   (b) A computer-generated image has the intensity distribution shown in the second column of the following table. 6

r_k	$p_r(r_k)$	Code 1	Code 2
$r_{87} = 87$	0.25	01010111	01
$r_{128} = 128$	0.47	01010111	1
$r_{186} = 186$	0.25	01010111	000
$r_{255} = 255$	0.03	01010111	001

Now, determine the average number of bits –

(i) if a natural 8-bit binary code (denoted as code 1 in the table) is used,

(ii) if the scheme designated as code 2 in the table is used to represent its four possible intensities.

(iii) For Code 2, find out the compression ratio.

2. (a) Explain objective fidelity criteria with some examples. 3

   (b) Consider the following source symbols and their probabilities: 5

Original Source	
Symbol	Probability
a_1	0.15
a_2	0.25
a_3	0.08
a_4	0.35
a_5	0.02
a_6	0.10
a_7	0.05

(i) Apply the Huffman Coding algorithm to construct the Huffman tree and determine the binary codeword for each symbol.

(ii) Determine the average length of the Huffman code.

(iii) Using the Huffman codes obtained in part (i), decode the encoded string 10100100100011 by performing a left-to-right scan.

3. (a) Differentiate between lossless and lossy compression system. 2

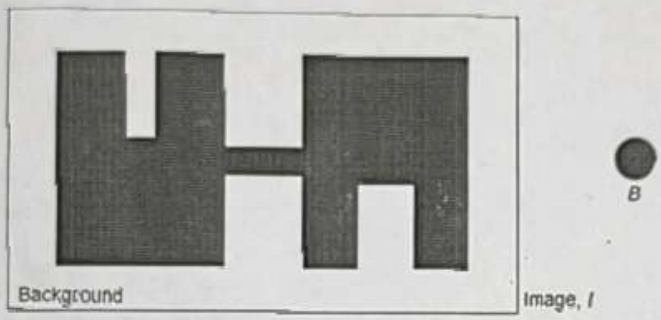
(b) Consider the following 3×3, 8-bit image of a vertical edge: 6

40 40 200
40 40 200
200 200 40

Encode the pixel sequence using LZW coding, obtained by scanning the image row by row and show all dictionary entries added. Assume the initial dictionary contains all individual pixel values as single-symbol entries.

4. (a) Explain structuring element with some examples. 2

(b) Consider the following image I with the structuring element B. Show the final effect after performing opening and closing operations on I using B. 6



5. (a) "The output shape of a morphological operation depends on the structuring element" – justify this statement with suitable examples. 4

(b) Explain the concept of boundary extraction in digital image processing and state the morphological equation used to extract the boundary of a binary image. 4

6. (a) State the key conditions that must be satisfied when partitioning an image's spatial region R into sub regions $R_1, R_2, \ldots, R_n$ during image segmentation. 2

(b) Explain the differences between discontinuity-based and similarity-based segmentation. 3

(c) Why is noise reduction necessary for edge detection in image segmentation? Explain. 3

7. (a) Briefly describe quantitative and structural patterns in image pattern recognition. 3

(b) Explain the basic architecture of a Convolutional Neural Network (CNN) and describe how it operates for image pattern recognition. 5